Epidemiology and Global Health

A natural experiment in Kenya reveals durable immunosuppressive effects of early childhood malaria: a longitudinal cohort study

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eLife Assessment

This **important** study sought to investigate the role that early childhood malaria exposure plays in the development of antibody responses to unrelated pathogens and vaccine-derived antigens in Kenyan children. In this natural experiment, the authors compare antibody levels among children who have been exposed to different levels of malaria transmission by using protein microarray technology. Although the findings are of importance, the evidence remains **incomplete**, and the analysis would benefit from a more in-depth evaluation of potential confounders. With the appropriate analysis, the findings will be of great interest for global health, immunology, and vaccine development.

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Abstract

Background

Chronic malaria exposure has been proposed to modulate immune function, but its long-term effects on antibody-mediated responses to unrelated pathogens remain poorly defined. Whether these effects persist beyond periods of active infection, and how early-life exposure shapes humoral immunity over time, is not well understood.

Methods

We leveraged a natural experiment in coastal Kenya - where two regions (Junju and Ngerenya) diverged sharply in malaria transmission from around 2004 - to evaluate the long-term immunological consequences of malaria exposure in childhood. Using a protein microarray platform, we measured IgG responses to vaccine and pathogen antigens in 123 children sampled longitudinally over a 15-year period. Active weekly malaria surveillance enabled precise reconstruction of individual exposure histories.

Results

IgG responses to *Plasmodium falciparum* apical membrane antigen 1 (AMA1) tracked closely with clinical malaria episodes, confirming the ability of the microarray platform to detect biologically meaningful variation in antigen-specific immunity. Despite comparable vaccination histories, children from the high malaria transmission setting (Junju) exhibited persistently lower measles-specific IgG levels than children from the low-transmission setting (Ngerenya), a pattern validated by ELISA. At 10 years of age, Junju children showed significantly reduced antibody levels to a wide range of unrelated pathogens, including *Bordetella pertussis*, CMV, rubella, and measles. Within the Ngerenya cohort, children with documented early-life malaria had broadly lower IgG responses at age 10 compared to malaria-naïve peers, despite identical geography, vaccines, and follow-up duration.

Conclusions

These findings suggest that malaria exposure during early childhood is linked with durable suppression of antibody responses to unrelated pathogens and vaccines. This effect persists long after infection and may partially explain the overall diminished long-term vaccine effectiveness in malaria-endemic settings.

Introduction

Malaria remains a major cause of childhood morbidity and mortality in sub-Saharan Africa. In 2023, Africa accounted for 95% of global malaria deaths, with over three-quarters occurring in children under five years old[1]. Alongside malaria, children in these settings face high exposure to a wide array of pathogens - including respiratory, enteric, and helminth infections [2], making immune competence critical for survival and long-term health. Despite widespread vaccination efforts, several studies report reduced vaccine efficacy in malaria-endemic regions compared to malaria-naïve populations [3–5]. Although multiple factors may contribute[6], growing evidence suggests that malaria itself can impair host immunity[5,7,8]. Repeated P. falciparum infection has been associated with immunomodulatory changes, including expansion of regulatory T cells [9,10], regulatory B cells[11], and atypical memory B cells with limited effector function[12,13]. These alterations promote host tolerance for persistent parasitaemia, resulting from recurrent exposure in endemic settings, but they also suppress effector immune responses[14], potentially to unrelated antigens, including vaccines. While malaria-induced immunosuppression has been described in both experimental and observational studies[7,8,15,16], its duration and broader impact on the developing immune system remain poorly understood. Most prior work has focused on short-term or antigen-specific outcomes, leaving open the question of whether early-life malaria exposure durably attenuates antibody responses over the long term. Conflicting findings across settings highlight the need for context-specific, and longitudinal data studies[16,17].

Here, we address this gap using two longitudinal cohorts from coastal Kenya with contrasting malaria transmission histories. One region (Junju) has experienced a sustained malaria burden, while the other (Ngerenya) underwent a rapid decline in transmission after 2004[18]. Intensive weekly malaria surveillance was conducted over more than a decade, enabling precise classification of individual exposure histories. We combined these data with serial serological measurements using a high-throughput in-house protein microarray. This design enabled us to investigate whether clinical malaria in early life leaves a lasting immunological imprint that compromises antibody responses to common childhood pathogens and vaccines. Our findings reveal a durable suppression of humoral immunity linked to malaria exposure in early childhood, with implications for vaccine effectiveness, serosurveillance interpretation, and immune recovery in endemic regions.

Materials and methods

Study setting and cohort design

This study was conducted in Kilifi County, a rural region on the northern coast of Kenya, within the catchment area of the Kilifi Health and Demographic Surveillance System (KHDSS), a long-term population-based platform maintained by the KEMRI-Wellcome Trust Research Programme[19]. Serum samples were obtained retrospectively from two well-characterised paediatric cohorts - Ngerenya and Junju - enrolled between 1998 and 2017 as part of annual malaria cross-sectional surveys[20]. Ngerenya and Junju were selected for their contrasting malaria transmission intensities; Ngerenya experienced a sharp decline in malaria transmission beginning in the early 2000s[18], whereas Junju maintained moderate malaria endemicity throughout the study period, with *P.falciparum* prevalence approximating 30% during the rainy seasons[21]. All children were visited weekly at home for the detection of febrile episodes, and any child with an axillary temperature ≥37.5°C was tested for P. falciparum parasitaemia using a rapid diagnostic test, with confirmation by microscopic examination of Giemsa-stained thick and thin blood smears. A clinical malaria episode was defined as fever in the presence of ≥2,500 parasites/µL. In addition to active malaria surveillance, serum samples were collected annually from each child and for future serological analysis. The vaccination status of each child for routine childhood vaccines was assessed using digitised immunisation records stored at the KEMRI-Wellcome Trust Research Programme.

Protein microarray antibody profiling

Antibody responses were measured using an in-house protein microarray platform. Recombinant and whole-virus lysate antigens (**Supplementary table 1**) were reconstituted in a glycerol-based buffer containing 1% Triton X-100 and printed in quadruplicate onto epoxy-coated glass slides using a non-contact microarrayer (Marathon Argus, Arrayjet, Scotland). Slides were fitted into hybridisation cassettes, washed with PBST (0.05% Tween-20 in PBS), and blocked for 1 hour at 37°C using PBST containing 5% BSA. Serum samples were diluted 1:30 in PBST with 5% BSA and incubated on the slides for 3 hours at room temperature. Slides were washed and probed with secondary antibodies: goat anti-human IgG conjugated to Alexa Fluor 647 (SouthernBiotech, cat. no. 2040-03) and goat anti-human IgA conjugated to Alexa Fluor 555 (SouthernBiotech, cat. no. 2050-32), each incubated for 1.5 hours at 25°C. Slides were rinsed, disassembled, and scanned using a GenePix 4300A scanner (635 nm at PMT 600, 532 nm at PMT 700). Mean fluorescence intensities (MFIs) were extracted from duplicate mini-arrays per slide and averaged across four replicate spots per antigen. Data processing and quality control of microarray data were performed in R (version 4.4.2).

Enzyme-linked immunosorbent assay (ELISA)

Measles-specific IgG was quantified using a conventional direct ELISA. Plates were coated overnight at 4°C with 2.30 µg/mL measles antigen diluted in PBS. After blocking with 5% skimmed milk for 1 hour at 37°C, serum samples (1:100 dilution) were added and incubated for 1.5 hours at 37°C. Plates were washed with PBST and incubated with HRP-conjugated secondary antibody (1:100 dilution) for 1 hour. Following additional washes, 100 µL of OPD substrate solution (30 mg OPD in 30 mL PBS with 30 µL H_2O_2) was added and incubated in the dark for 10 minutes. Reactions were stopped with 50 µL of 2.5 M sulfuric acid, and absorbance was measured at 490 nm.

Statistical analysis

All statistical analyses were performed using R (version 4.4.2). Antibody responses between cohorts were compared using the Wilcoxon rank-sum test. P-values <0.05 were considered statistically significant. Data were anonymised and delinked from all personally identifiable information.

Ethics statement

The study was approved by the Scientific and Ethics Review Unit (SERU) of the Kenya Medical Research Institute. All procedures were conducted in accordance with the principles of Good Clinical Laboratory Practice (GCLP).

Results

A natural experiment in coastal Kenya reveals sharply divergent malaria exposure trajectories in early childhood

Between 1998 and 2017, two rural communities in coastal Kenya - Junju and Ngerenya - underwent markedly different transitions in malaria transmission. Both regions experienced a high malaria burden in the 1990s and early 2000s. However, beginning in 2004, transmission in Ngerenya declined sharply and remained near zero for more than a decade, while Junju continued to experience sustained endemicity well into the mid-2010s. To quantify the scale and timing of this transition, we analysed surveillance data collected from August 1998 to April 2017. A total of 1,243 children were followed in Ngerenya and 659 in Junju. In Ngerenya, the proportion of febrile surveillance visits with confirmed P. falciparum parasitaemia declined from 22.4% before 2004 (1,378 of 6,148 visits) to just 1.1% after 2004 (48 of 4,389 visits). In contrast, Junju children continued to experience high rates of febrile malaria, with 17% of visits testing positive between 2007 and 2017 (869 of 5,130 visits) - **Fig. 1**.

To assess whether the protein microarray platform accurately captured longitudinal P. falciparum-specific immune responses, we examined IgG levels against the plasmodium apical membrane antigen 1 (AMA1) in individual children with known malaria exposure histories. Among children with multiple confirmed episodes of febrile malaria detected via weekly active surveillance, AMA1-specific IgG levels increased sharply over time, tracking closely with the timing of clinical infections (example shown in **Fig. 2a**), while children from Ngerenya who remained malaria-free throughout follow-up exhibited consistently low IgG levels against AMA1 over more than a decade of surveillance (example shown in **Fig. 2b**). These patterns mirrored expected immunological trajectories of repeated exposure versus non-exposure, confirming that the microarray platform is capable of detecting meaningful variation in P. falciparum-specific humoral responses. We then extended this analysis to assess longitudinal IgG responses to AMA1 in a subset of 123 children drawn from the two surveillance cohorts. These children contributed a total of 1,194 serum samples, with a mean of 10 samples per child, collected between 2002 and

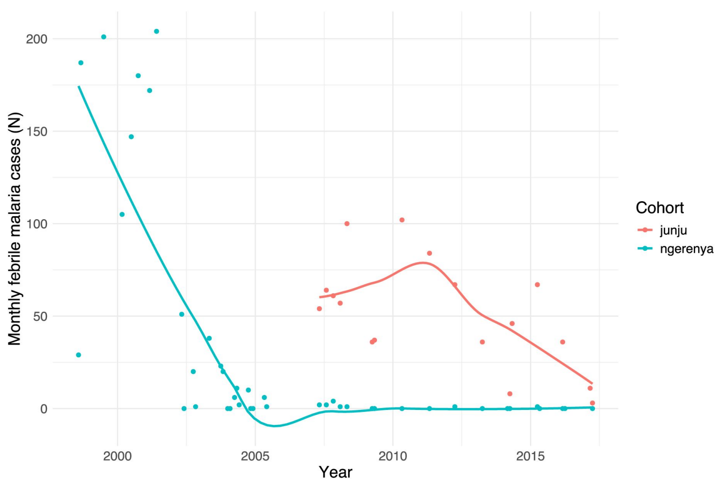

Figure 1

Divergent malaria exposure histories in coastal Kenya

Monthly malaria case counts from active surveillance in two adjacent regions of Kilifi County, Kenya, between 1998 and 2017. Junju (red) maintained sustained malaria transmission throughout the study period, while Ngerenya (blue) experienced a rapid collapse in transmission beginning in 2004. Points represent total cases per month; lines show LOESS-smoothed trends.

2017 (**Supplementary Fig. 1**). This subset was selected on the basis of the relative completeness of their longitudinal follow-up serum sampling and included 58 children from Junju (505 samples) and 65 from Ngerenya (689 samples). AMA1-specific IgG levels diverged sharply between cohorts early in life and remained distinct throughout follow-up. In Ngerenya, levels declined rapidly after 2003, stabilising at lower levels by mid-childhood. In contrast, Junju children maintained elevated levels across all time points (**Fig. 2c**).

Malaria-exposed children exhibit lower antibody levels to non-malarial antigens

To examine the impact of differential malaria endemicity on the antibody response to non-malarial antigens, we first compared IgG responses to the measles virus among Junju and Ngerenya children. We started by validating the ability of the in-house protein microarray platform to detect biologically meaningful measles-specific IgG responses by examining longitudinal measles levels in individual children with known vaccination histories. Among children with a documented history of receiving all three recommended doses of measles vaccine, we observed a sharp rise in IgG levels following immunisation, followed by sustained levels into later childhood (example shown in **Fig. 3a**). In contrast, unvaccinated children exhibited consistently low IgG trajectories across all time points (example shown in **Fig. 3b**). These patterns were reproducible across the cohort and recapitulated expected vaccine-induced versus naïve antibody dynamics, supporting the use of this platform for population-level serological inference. Using this platform, we then compared IgG responses to measles virus among children from Junju and Ngerenya. This analysis was restricted only to children with a documented record of measles vaccination. Despite matched vaccination histories, children from Junju - where malaria transmission remained high - consistently exhibited lower measles-specific IgG titres than children from Ngerenya, where malaria transmission had declined. This difference was evident from the earliest timepoints and persisted throughout childhood. Annual antibody measurements showed that mean measles-specific IgG levels were higher in Ngerenya than in Junju in every sampling year (**Fig. 3c**). To validate these findings, we measured measles-specific IgG levels by ELISA in a subset of 3-year-old children who had completed measles vaccination at least one year prior. In this assay, Junju children again displayed significantly lower IgG levels than their Ngerenya counterparts (**Fig. 3d**).

Early-life malaria exposure is associated with long-term suppression of antibody responses

To assess whether the attenuation of antibody responses extended beyond measles, we compared IgG levels to a broader panel of antigens at 10 years of age. This analysis included vaccine-preventable pathogens (*Bordetella pertussis*, H1N1 influenza virus, rubella virus, and measles virus) and common childhood infections (cytomegalovirus (CMV), Epstein-Barr virus (EBV), herpes simplex virus 1 (HSV-1), coxsackievirus B1). Children from Junju exhibited significantly lower IgG levels for most antigens compared to their Ngerenya counterparts (**Fig. 4**). This pattern was observed for both viral and bacterial pathogens and was particularly marked for coxsackievirus, EBV, HSV-1, and measles. Antibody levels against P. falciparum AMA1 remained elevated in Junju. Antibody levels to the 2009 pandemic strain of H1N1 were the least differentiated between cohorts, although Ngerenya children still exhibited significantly higher levels than Junju participants (**Fig. 4**).

To determine whether the attenuation of antibody responses could still be attributed to early-life malaria exposure independent of geographic or environmental differences, we conducted a stratified analysis within the Ngerenya cohort, which experienced a sharp decline in malaria transmission beginning in 2004. Due to stochastic differences in malaria infection around the time of this inflection, children in Ngerenya had different malaria exposure histories despite living in

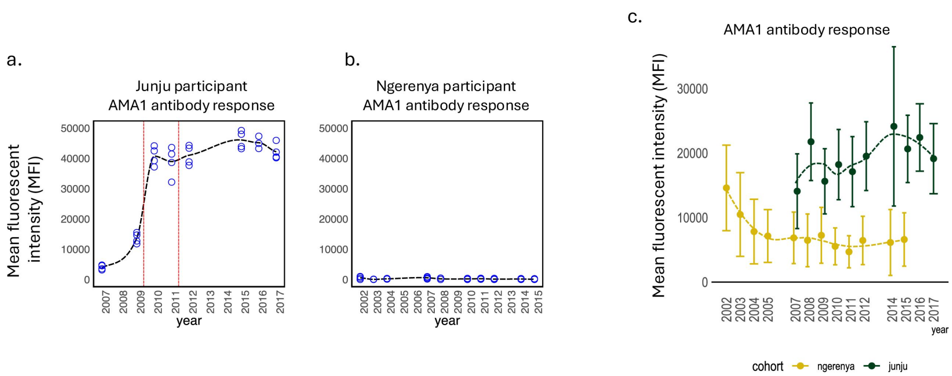

Figure 2

AMA1-specific IgG trajectories mirror individual and regional malaria exposure

(A–B) Longitudinal AMA1 IgG levels in individual children measured by protein microarray. Vertical red lines denote confirmed febrile malaria episodes. Panel A shows a Junju child with multiple documented infections; Panel B shows a Ngerenya child who remained malaria-free throughout follow-up. Each blue spot is a single antibody measurement. Each time point was measured in quadruplicate **(C)** Mean AMA1-specific IgG levels with 95% confidence intervals for all children in the microarray subset plotted by year of sampling. Junju children showed persistently elevated antibodies, while AMA1 antibody levels in Ngerenya declined sharply after 2004.

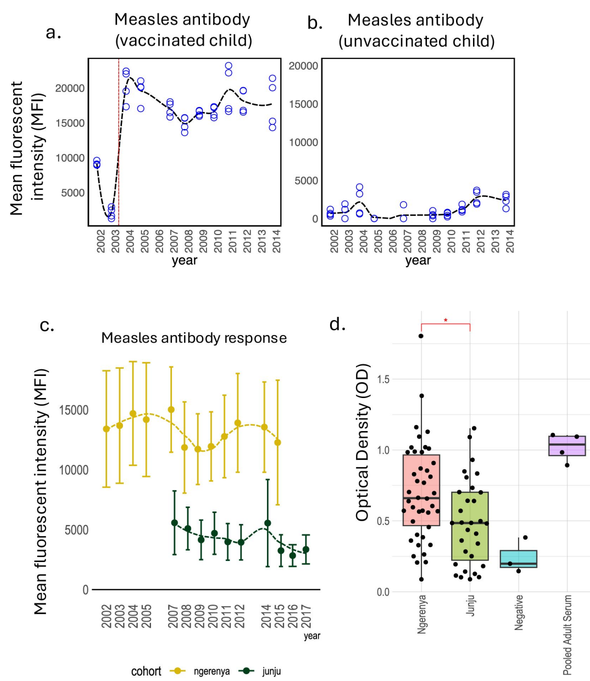

Figure 3

Malaria exposure is associated with reduced measles-specific antibody levels.

Example plots of temporal changes in measles-specific antibody in vaccinated and unvaccinated children are shown **(A)** Longitudinal IgG levels in an individual child measured by microarray. The dashed vertical red line indicates the timing of the last dose of the routine measles vaccine. **(B)** Shows a similar temporal trend for a child with no history of measles vaccination. **(C)** Longitudinal IgG responses to measles virus by cohort, measured by protein microarray in vaccinated children. Junju children exhibited consistently lower levels of measles-specific antibody than Ngerenya counterparts. The circles indicate means, and the whiskers denote 95% confidence intervals. **(D)** Measles-specific IgG levels in 3-year-old children from Junju and Ngerenya, measured by ELISA in children with a documented history of measles vaccination. Each dot represents an individual participant.

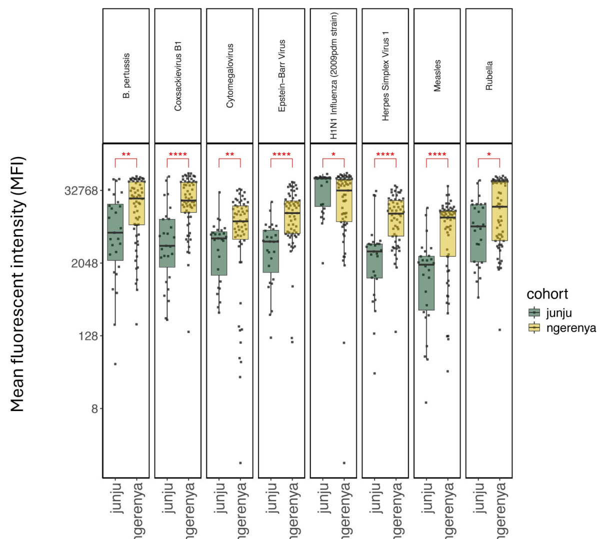

Figure 4

Broad suppression of antibody responses to diverse pathogens in malaria-exposed children

cross-sectional IgG titres at 10 years of age for a panel of pathogen and vaccine antigens, measured by protein microarray. Each dot represents an individual child; boxes indicate interquartile ranges, with the midline denoting the median. Junju children show significantly lower antibody levels across most antigens compared to Ngerenya children, including *Bordetella pertussis*, H1N1 influenza virus, rubella virus, measles virus, cytomegalovirus (CMV), Epstein-Barr virus (EBV), herpes simplex virus 1 (HSV-1), and coxsackievirus B1.

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the same area. At the 10-year-of-age time point, sera were available for 62 out of the 65 children that were originally selected in the Ngerenya cohort subset for serological analysis. Of these, 20 experienced one or more episodes of febrile malaria during early childhood prior to the decline in malaria transmission, while 42 remained entirely malaria-free throughout follow-up (**Fig. 5a**). At 10 years of age, Ngerenya children who had experienced early-life malaria exhibited significantly lower IgG levels to a wide range of antigens compared to their malaria-naïve peers (**Fig. 5b**). These included responses to Coxsackievirus B1, CMV, H1N1 influenza, HSV-1 and rubella. Differences in antibody level to *B. pertussis* and EBV did not reach statistical significance.

Discussion

This study demonstrates that early-life exposure to malaria is associated with broad and durable impairments in antibody-mediated immunity to unrelated pathogens and vaccines. By leveraging a natural experiment in coastal Kenya - where malaria transmission diverged sharply between adjacent communities during the early 2000s - we were able to disentangle the immunological effects of malaria exposure from confounding geographic and temporal factors. Our findings reveal that children exposed to malaria in early childhood not only generate lower antibody titres to non-malarial antigens but maintain these attenuated responses well into adolescence, long after malaria transmission has ceased. The attenuating effect of malaria on antibody responses has long been suspected [15] but difficult to quantify. Prior studies have shown reduced vaccine efficacy in malaria-endemic settings [3,16], and experimental models have implicated regulatory T cells, atypical memory B cells, and B cell exhaustion as potential mediators of malaria-induced immune suppression[9,12,13,22]. However, most human studies to date have focused on short-term immune responses or outcomes in the setting of concurrent parasitaemia. Our data extend this work by demonstrating that transient exposure to malaria in early childhood is sufficient to imprint long-lasting changes on the humoral immune repertoire.

The strength of this study lies in its integration of long-term active malaria surveillance with serial antibody profiling. Children were visited weekly for febrile illness surveillance, allowing for precise documentation of clinical malaria episodes. In parallel, our use of a validated protein microarray platform enabled us to track IgG responses to a wide panel of pathogen and vaccine antigens at multiple timepoints. We first confirmed that the microarray platform captured biologically relevant responses by comparing antibody trajectories in individual children with known measles vaccination and malaria exposure histories. The platform reliably detected both vaccine-induced and infection-associated rises in IgG, providing confidence in the longitudinal patterns observed. We found that children from Junju, a region of persistent transmission, had significantly lower antibody levels to multiple pathogens compared to children from Ngerenya, where malaria transmission declined in the mid-2000s. These differences were most pronounced for antigens such as measles virus, *Bordetella pertussis*, CMV, rubella virus, and HSV-1. Importantly, all children had comparable vaccination histories and were followed through the same longitudinal infrastructure, minimising the likelihood of differential healthcare access or vaccine uptake.

To test whether early-life malaria exposure specifically contributed to long-term immune suppression, we examined antibody responses within the Ngerenya cohort. Because the transmission decline occurred rapidly, children born just before and after the inflection point experienced markedly different levels of malaria exposure while living in the same geographic area. Among children followed longitudinally, those with even limited early-life exposure to malaria had significantly lower antibody titres at 10 years of age than their malaria-naïve peers. This within-cohort contrast strongly implicates early-life infection as the critical window for immune programming. Taken together, these findings support a model in which malaria exposure during critical developmental windows modulates long-term maintenance of immune memory. This may occur through direct effects on B cell maturation, altered antigen presentation[23], or

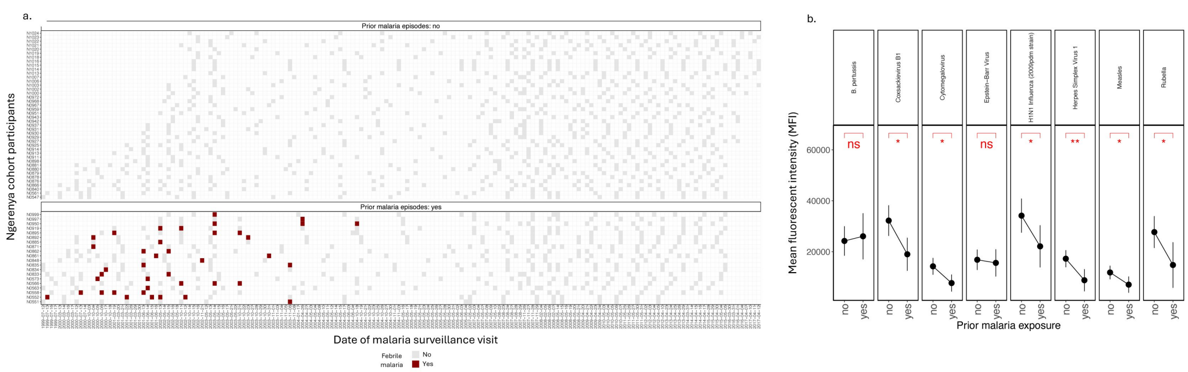


Figure 5

Early-life malaria exposure predicts long-term suppression of antibody responses within the same geographic region

(A) Active malaria surveillance records for children in the Ngerenya cohort. Each row represents an individual child, and each column represents a surveillance timepoint. Dark red boxes indicate one or more confirmed febrile malaria episodes; light grey boxes indicate surveillance visits without malaria detection. Children are grouped by early-life exposure status (top: malaria-naïve; bottom: previously exposed). **(B)** IgG levels at 10 years of age among Ngerenya children, stratified by early-life malaria exposure. Children with ≥1 confirmed febrile malaria episode during early childhood (n = 20) show significantly lower titres to multiple unrelated pathogens compared to malaria-naïve peers (n = 42). All children lived in the same geographic area and received identical vaccines and follow-up. The black dots are means and error bars are 95% confidence intervals.

long-lived changes in lymphoid microenvironments[24–26]. While the mechanistic basis of this phenomenon remains to be fully explained, the consistency of the attenuation on multiple antigens and timepoints suggests a generalizable effect rather than antigen-specific phenomenon.

This work has important implications for vaccine policy and infection risk in malaria-endemic regions. Reduced antibody titres to vaccine-preventable diseases may translate into diminished long-term protection, even when vaccine coverage is high. Our findings raise the possibility that children in high-transmission settings may require different immunisation strategies, such as delayed dosing and boosting, to improve vaccine-induced immunity. Moreover, as malaria control efforts continue to shift disease epidemiology, these data highlight the value of longitudinal serological surveillance for understanding the broader immunological legacy of malaria exposure. Our study has several limitations. Although the microarray platform was validated internally and against ELISA, quantitative comparisons across platforms are inherently constrained. Additionally, while the sample size for antibody profiling was modest, the inclusion of dense longitudinal sampling and detailed clinical surveillance adds depth to the dataset. Finally, we cannot fully exclude the influence of other infections or environmental exposures that may have differed subtly between subgroups, although the within-Ngerenya analysis provides strong evidence for malaria as a primary driver of long-term immune attenuation.

In summary, our findings reveal that early-life malaria exposure is associated with long-term suppression of antibody responses to unrelated pathogens and vaccines. This effect is detectable many years after infection, and appears to persist even in the absence of ongoing transmission. As global malaria control efforts continue, understanding the immunological legacy of childhood malaria may be critical for improving vaccination strategies and mitigating susceptibility to other infections.

Data and Code Availability

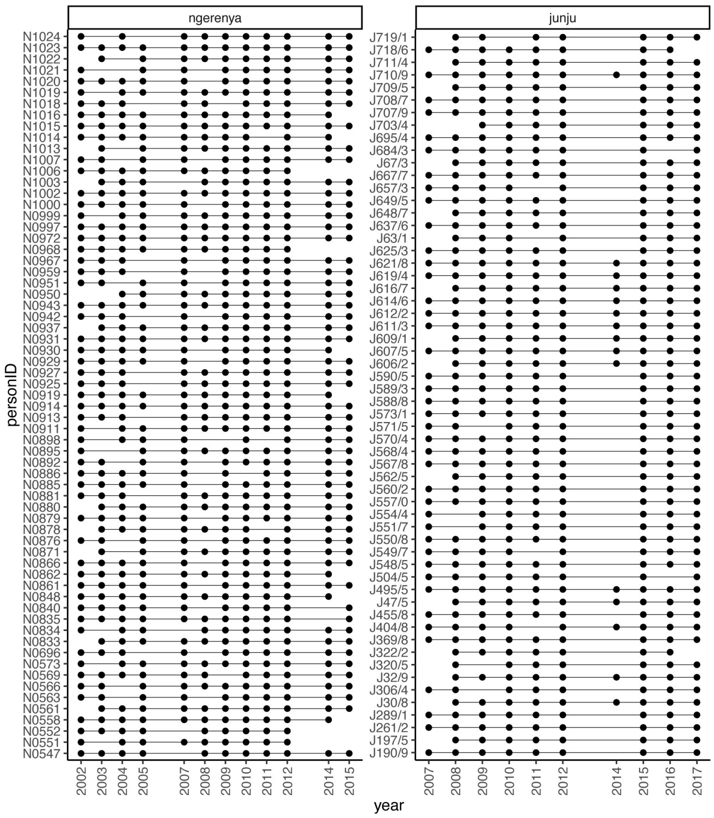
The datasets and R scripts used for the analysis in this study are available from the corresponding author upon reasonable request. Due to local laws and regulations, individual-level data from the Kilifi Health and Demographic Surveillance System (KHDSS), cannot be made publicly available. All data processing and visualisation were performed using open-source R packages; package versions and analysis pipelines are available upon request.

Supplementary Materials

Supplementary Figure 1

A summary of the sampling frame for the study cohorts (Junju and Ngerenya).

Each vertical line represents the longitudinal time series for a single individual, and each dot represents a timepoint where a serum sample was collected.



Supplementary Table 1

Summary of pathogens, strains or subtypes, and corresponding antigens included in the immunoassay panel.

Pathogen	Strain/subtype	Antigen
Measles virus	Edmonston	Whole virus antigen
Coxsackie B virus	B1	VP1
Cytomegalovirus	TB40E	PP150
Epstein-Barr virus	Type 1	Nuclear antigen 1
Herpes simplex virus	Type 1	Extract
H1N1 Influenza A	A/California/07/2009	Haemagluttinin
Plasmodium falciparum	3D7	AMA-1
Rubella virus	HPV-77	Whole virus antigen

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Reviewer #1 (Public review):

Summary:

The study shows that childhood malaria can weaken the antibody response to other vaccines and infections. This suggests that early exposure to P. falciparum may have a long-lasting effect on immunity, with implications for vaccine efficacy in endemic areas.

Strengths:

This study stands out for its longitudinal design, the use of robust immunological techniques, and the comparison between areas with different levels of malaria exposure. Its findings reveal that early malaria can weaken the response to childhood vaccines, with important implications for public health in endemic regions.

Weaknesses:

One of the study's main limitations is the lack of functional data confirming the clinical impact of the low antibody levels. Furthermore, although multiple immune responses were measured, other important components, such as cellular immunity, were not assessed. Furthermore, the results may not be generalizable to other regions.

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Reviewer #2 (Public review):

Summary:

The authors investigated whether early-life malaria exposure has long-term effects on immune responses to unrelated antigens. They leveraged a natural experiment in coastal Kenya where two adjacent communities (Junju and Ngerenya) experienced divergent malaria transmission patterns after 2004. Using 15 years of longitudinal data from 123 children with weekly malaria surveillance and annual serological sampling, they measured antibody responses to multiple pathogens using a protein microarray technology and ELISA.

Strengths:

(1) Extensive longitudinal data collection with weekly malaria surveillance, enabling precise exposure classification.

(2) Use of a natural experiment design that allows for causal inference about malaria's immunological effects.

(3) Broad panel of antigens tested, demonstrating generalized rather than antigen-specific effects.

(4) Within-cohort analysis in Ngerenya controls for geographic and environmental factors.

(5) Validation of key findings using both serologic microarray and ELISA.

(6) Important public health implications for vaccine strategies in malaria-endemic regions.

Weaknesses:

(1) Lack of participants' characteristics (socio-economic, nutritional, physical).

(2) Somewhat limited sample size (longitudinal analysis of 123 children total), with further subdivision reducing statistical power for some analyses.

(3) Potential confounding by unmeasured socioeconomic, nutritional, or environmental factors between communities.

(4) Lack of ability to determine the direction of the associations found between malaria exposure and other IgG levels to unrelated pathogens.

(5) Despite good longitudinal data, the main analysis was conducted as a cross-sectional analysis at age 10 for many comparisons, which limits the understanding of temporal dynamics.

(6) Statistical analysis is limited to univariable comparisons without consideration for confounders or adjusting for multiple comparisons.

(7) No mechanistic understanding of how early malaria exposure creates lasting immunosuppression.

(8) No understanding of the clinical Implications of the reduced IgG levels observed in the area with high malaria exposure.

Assessment of Claims:

The data appear to support the authors' primary claims, but the strength of the evidence is limited, and the results should be interpreted with caution. Together with the currently available evidence of P. falciparum's impact on the host's immune function, this natural experiment design provides further evidence for a relationship between early malaria exposure and reduced antibody responses. The within-Ngerenya analysis controls for geographic factors and thus enhances the quality of the evidence; however, it still fails to account for the physical, nutritional, and socio-economic factors that may have driven the observed changes. Additionally, the mechanism underlying this effect remains unclear, and the clinical significance of reduced antibody levels is not established.

Impact and Utility:

This work has fundamental implications for understanding vaccine effectiveness in malaria-endemic regions and may contribute to informing vaccination strategies. The findings, if strengthened, would suggest that children in areas of high malaria transmission may require modified immunization approaches. The dataset provides a valuable resource for future studies of malaria's immunological legacy.

Context:

This study builds on prior work showing acute immunosuppressive effects of malaria but uniquely attempts to demonstrate the durability of these effects years after exposure. The natural experiment design addresses limitations of previous observational studies by providing a more controlled comparison.

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